

# Baptiste Girardin

Paris, France · +33 7 82 82 91 83 · [baptiste.girardin@telecom-paris.fr](mailto:baptiste.girardin@telecom-paris.fr) · [linkedin.com/in/baptiste-girardin](https://linkedin.com/in/baptiste-girardin) · [github.com/SlyZ1](https://github.com/SlyZ1) · [slyz1.github.io](https://slyz1.github.io) (portfolio)

## EDUCATION

### Télécom Paris • Engineering School (MSc of Engineering) ·

Sep. 2024 - Present

Institut Polytechnique de Paris • Top French engineering school

Tracks: 3D & Interactive Systems, Computer Vision & Artificial Intelligence

### Preparatory Classes for French Grandes Ecoles (MPSI/MP) ·

2022 - 2024

Lycée Camille Guérin • Poitiers

- Ranked top of class both years.
- Intensive program in Mathematics, Physics, Computer Science.

## EXPERIENCE

### Industrial Internship ·

Jul. - Aug. 2025

SKF • Saint-Cyr-sur-Loire

- Reorganized and restructured digital file systems across departments
- Automated email workflows to streamline internal communications

### Board Member - La Scène (Live Music Student Association) ·

2025 - 2026

Télécom Paris

- Designed promotional materials (event posters, merchandise) and managed ordering and sales
- Helped organize music festivals (300+ attendees)
- Performed in a jazz band at association events

## PROJECTS

### Spectral & RGB Path Tracer ·

2026 - Now

[github.com/SlyZ1/Path-Tracer](https://github.com/SlyZ1/Path-Tracer)

Interactive physically based renderer built with OpenGL and c++.

- MIS, NEE, BVH for SAH heuristic, EON diffuse, VNDF sampling, artistic fresnel for conductors.
- Spectral rendering, Hero-wavelength sampling, dielectric dispersion, thin-film iridescence.
- Physically based hair/fur scattering (Chiang et al.), artist-friendly roughnesses.

### Real-Time Smoke Simulation & Rendering ·

2025

[github.com/artishow-smoke/Smoke](https://github.com/artishow-smoke/Smoke)

Group project in Godot (C# & HLSL). Fluid simulation via LBM and ray-marching for single scattering.

### Soft & Rigid Body Physics Simulation ·

2026

[github.com/SlyZ1/Physics-Based-Bodies](https://github.com/SlyZ1/Physics-Based-Bodies)

Group project in Godot simulating real time soft-rigid body interactions using spring physics and soft constraints.

### Face Edition by Latent Code Manipulation ·

2026

[github.com/SlyZ1/Latent-Space-Faces](https://github.com/SlyZ1/Latent-Space-Faces)

Group project experimenting with StyleGAN2-ADA pre-trained models to generate and manipulate photorealistic faces (Python, PyTorch).

### Echo Depths • Ludum Dare 57, ranked ~50th / 3 000 entries ·

2025

[github.com/Protocol-Juice/echo-depths](https://github.com/Protocol-Juice/echo-depths)

48-hour game jam entry built on Unity with two teammates. Full game design, art, and programming.

### Isometric Procedural Environment ·

2024

[github.com/SlyZ1/Isometric-Pixel-Game](https://github.com/SlyZ1/Isometric-Pixel-Game)

Isometric pixel-art game in Unity with procedural generation and a custom lighting system.

## SKILLS

**Languages:** C++, C#, C, Python, PyTorch, GLSL / HLSL

**Tools & Engines:** OpenGL, Unity, Godot, Blender, Git, LaTeX, Typst

**Human languages:**

French (native), English (C2)

## INTERESTS

Computer graphics · Piano & jazz improvisation · Game development · Rock climbing · Graphic design